

MEMSYST STAGE 13 DEVELOPMENT MASTER PROMPT

Analytics, Intelligence, Reporting & Decision Support Platform (AIRDSP)

You are a Senior Data Platform Architect, Business Intelligence Architect, Enterprise Analytics Architect, Membership Technology Architect, and Principal Engineer responsible for building Stage 13 of the MemSyst ecosystem.

Stages 1–12 have already established:

Stage 1

Business Operations Platform

Stage 2

Identity & Access Platform

Stage 3

Tenant Management Platform

Stage 4

Membership & Apprentice Platform

Stage 5

Financial Infrastructure Platform

Stage 6

Communication & Engagement Platform

Stage 7

Digital Identity & Credential Platform

Stage 8

Marketplace, Business Directory & Commerce Platform

Stage 9

Training, Learning, Apprenticeship & Skills Development Platform

Stage 10

Support Hub, Opportunities, Grants, Loans & Member Services Platform

Stage 11

Events, Conferences, Meetings & Attendance Management Platform

Stage 12

Governance, Elections, Voting & Decision Management Platform

Stage 13 introduces the intelligence layer of the MemSyst ecosystem.

This stage transforms operational data into actionable insight, enabling leaders, executives, administrators, and MemSyst platform operators to make informed strategic decisions.

STAGE 13 OBJECTIVE

Build the:

Analytics, Intelligence, Reporting & Decision Support Platform (AIRDSP)

Purpose:

Provide:

Operational Analytics
Executive Dashboards
Strategic Reporting
Performance Monitoring
Cross-Tenant Intelligence
Financial Intelligence
Membership Intelligence
Predictive Readiness
Decision Support

for both:

MemSyst Platform
Tenant Organizations

CRITICAL ANALYTICS PRINCIPLES

Principle 1

Analytics Is Multi-Level

Analytics must support:

Platform Level
Tenant Level
National Level

Regional Level
Branch Level
Department Level
Program Level
Individual Level

depending on permissions.

Principle 2

Data Ownership Must Be Preserved

Tenant users may only access analytics for their tenant.

MemSyst administrators may access platform-wide analytics.

Principle 3

Reports Must Be Configurable

No hardcoded reporting assumptions.

Organizations must be able to create:

Custom Dashboards
Custom Reports
Custom KPIs
Custom Metrics

Principle 4

Analytics Must Be Real-Time Where Possible

Critical metrics should update automatically.

Historical reporting must remain available indefinitely.

STAGE 13 DELIVERABLES

Build:

Enterprise Analytics Engine
Reporting Engine
Dashboard Engine

Data Warehouse Layer
KPI Engine
Insights Engine
Decision Support Engine
Executive Intelligence Engine

MODULE 1

ENTERPRISE ANALYTICS ENGINE

Purpose:

Central analytics infrastructure.

Data Sources

Collect data from:

Membership
Finance
Communications
Marketplace
Training
Support Hub
Events
Governance
Credentials
Tenant Management

Requirements

Analytics must aggregate:

Live Data
Historical Data
Trend Data
Comparative Data

MODULE 2

DATA WAREHOUSE LAYER

Purpose:

Provide analytical storage separate from transactional systems.

Requirements

Design:

Fact Tables

Dimension Tables

Aggregated Metrics

Historical Snapshots

Goals

Support:

Fast Reporting

Historical Analysis

Executive Dashboards

Future AI Analytics

MODULE 3

DASHBOARD ENGINE

Purpose:

Provide visual analytics.

Dashboard Types

Platform Dashboard

Tenant Dashboard

Executive Dashboard

Regional Dashboard

Branch Dashboard

Personal Dashboard

Features

Widgets
Charts
Tables
Scorecards
Trend Lines
KPIs

Requirements

Dashboard layouts must be configurable.

MODULE 4

MEMBERSHIP INTELLIGENCE

Purpose:

Provide membership insights.

Metrics

Total Members
Active Members
Inactive Members
New Members
Member Growth
Renewal Rates
Membership Categories
Member Retention
Apprentice Growth

Analysis

Monthly
Quarterly
Yearly
Custom Range

MODULE 5

FINANCIAL INTELLIGENCE

Purpose:

Provide financial visibility.

Metrics

Revenue
Collections
Wallet Balances
Outstanding Dues
Renewal Revenue
Transaction Volume
Commission Revenue
Messaging Revenue

Scope

Platform
Tenant
Region
Branch

MODULE 6

COMMUNICATION INTELLIGENCE

Purpose:

Measure engagement.

Metrics

Emails Sent
SMS Sent
Push Notifications Sent
Delivery Rates
Open Rates
Click Rates

Response Rates
Costs

Reporting

Track communication effectiveness over time.

MODULE 7

MARKETPLACE INTELLIGENCE

Purpose:

Measure economic engagement.

Metrics

Business Profiles

Marketplace Listings

Opportunities

Listing Views

Business Verification

Member Participation

Analysis

By Tenant

By Region

By Category

MODULE 8

TRAINING INTELLIGENCE

Purpose:

Measure learning performance.

Metrics

Enrollments
Course Completions
Graduations
Certifications
Pass Rates
Training Center Performance
Skills Achieved

MODULE 9

SUPPORT HUB INTELLIGENCE

Purpose:

Measure member support impact.

Metrics

Applications
Approvals
Loans
Repayments
Grants
Scholarships
Funding Utilization
Impact Scores

MODULE 10

EVENT INTELLIGENCE

Purpose:

Measure participation.

Metrics

Events
Registrations
Attendance
Ticket Revenue

Certificates Issued
Participation Rates

MODULE 11

GOVERNANCE INTELLIGENCE

Purpose:

Measure governance health.

Metrics

Election Participation
Voting Turnout
Committee Activity
Executive Terms
Resolution Activity
Governance Compliance

MODULE 12

KPI MANAGEMENT ENGINE

Purpose:

Allow organizations to define their own performance metrics.

Features

Create KPI
Edit KPI
Assign Targets
Track Progress
Alert Thresholds

Examples

Membership Growth
Revenue Growth
Renewal Rate

Training Completion Rate
Attendance Rate

MODULE 13

EXECUTIVE SCORECARD ENGINE

Purpose:

Provide leadership-level visibility.

Features

Organization Health Score
Membership Health
Financial Health
Engagement Health
Governance Health
Training Health

Output

Executive-ready dashboards.

MODULE 14

REPORT BUILDER ENGINE

Purpose:

Allow users to generate reports without technical skills.

Features

Drag-and-Drop Builder
Filters
Grouping
Sorting
Calculated Fields
Templates

Export Formats

PDF

Excel

CSV

MODULE 15

SCHEDULED REPORTING ENGINE

Purpose:

Automate reporting.

Features

Daily Reports

Weekly Reports

Monthly Reports

Quarterly Reports

Annual Reports

Custom Schedules

Delivery

Email

In-App

Download Center

MODULE 16

DECISION SUPPORT ENGINE

Purpose:

Provide actionable recommendations.

Examples

Declining Membership Alert
Low Renewal Alert
Revenue Risk Alert
Inactive Branch Alert
Training Performance Alert
Election Participation Alert

Requirements

Rule-based recommendations first.

Do not implement AI recommendations yet.

Prepare architecture for AI later.

MODULE 17

BENCHMARKING ENGINE

Purpose:

Enable comparative analysis.

Comparisons

Current Period vs Previous Period
Branch vs Branch
Region vs Region
Program vs Program
Tenant vs Tenant (Platform Admin Only)

Security

Tenant users must never see another tenant's detailed data.

Only aggregated benchmarking allowed where permitted.

MODULE 18

ANALYTICS API LAYER

Purpose:

Expose analytics services internally.

Consumers

Dashboards
Reports
Mobile Apps
Future AI Modules
Executive Portals

MODULE 19

DATA EXPORT ENGINE

Purpose:

Allow organizations to access their data.

Export Types

Membership Reports
Financial Reports
Training Reports
Event Reports
Governance Reports
Custom Reports

Formats

CSV
Excel
PDF

MODULE 20

ANALYTICS AUDIT ENGINE

Purpose:

Track analytics usage.

Audit Events

Report Generated
Dashboard Viewed
Data Exported
KPI Created
Scheduled Report Created

Record

User
Action
Date
Time
Filters Applied
Export Type

MODULE 21

ANALYTICS SETTINGS MANAGEMENT

Purpose:

Tenant-level analytics configuration.

Settings

KPIs
Dashboard Layouts
Report Templates
Benchmarks
Alert Rules
Scheduled Reports

Requirements

All settings configurable through UI.

No hardcoded reports.

DATABASE DESIGN REQUIREMENTS

Before implementation:

Design complete entities for:

Analytics Datasets

KPIs

Dashboards

Dashboard Widgets

Reports

Report Templates

Scheduled Reports

Benchmarks

Alerts

Exports

Analytics Audits

Define:

Relationships

Indexes

Aggregation Models

Retention Policies

Security Filters

Caching Strategies

SECURITY REQUIREMENTS

Integrate with Stage 2.

Enforce:

RBAC

Tenant Isolation

Report Permissions

Export Permissions

Dashboard Permissions

Audit Logging

PERFORMANCE REQUIREMENTS

Analytics must support:

- Large Datasets
- Thousands of Members
- Multiple Tenants
- Historical Analysis
- Concurrent Dashboard Users

Implement:

- Caching
- Aggregation Tables
- Background Processing
- Report Queues

FUTURE READINESS REQUIREMENTS

Stage 13 must prepare for:

- AI Analytics
- Predictive Analytics
- Machine Learning Models
- Natural Language Reporting
- Chat-Based Analytics
- Executive AI Assistants
- Forecasting
- Anomaly Detection

Do not build them now.

Design the architecture to support them later.

ARCHITECTURE REQUIREMENTS

Use:

- Data Warehouse Architecture
- Analytics Domain Architecture
- Event-Driven Data Collection
- Configurable Reporting Framework
- Audit-First Design

Avoid:

Direct Reporting Against Transactional Tables
Hardcoded Dashboards
Hardcoded KPIs
Tenant Data Leakage

SUCCESS CRITERIA

Stage 13 is complete only when:

- ✓ Enterprise analytics are operational
- ✓ Platform and tenant dashboards exist
- ✓ KPI management exists
- ✓ Executive scorecards exist
- ✓ Membership intelligence exists
- ✓ Financial intelligence exists
- ✓ Marketplace intelligence exists
- ✓ Training intelligence exists
- ✓ Support Hub intelligence exists
- ✓ Event intelligence exists
- ✓ Governance intelligence exists
- ✓ Report builder exists
- ✓ Scheduled reporting exists
- ✓ Benchmarking exists
- ✓ Decision support alerts exist
- ✓ Analytics audit history exists
- ✓ Platform is fully prepared for Stage 14 (Integration, API Gateway & Ecosystem Connectivity Platform)

Build this stage as the intelligence and decision-support infrastructure of the MemSys ecosystem, transforming operational data into measurable organizational insight and enabling data-driven leadership at every level of every tenant organization.